

SOFTWARE TEST PLAN (STP)

Event Management System

Project: Event Management system

Version: 1.0

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Revision History

Version	Date	Author	Change Summary	Approval
1.0	15-09-2026	Team	Initial Software Test Plan for Event Management System	
1.0	16/09/2026	Team Lead (Aaryesh)	Features to be tested and test approach and strategy, validation of report	
1.0	16/09/2026	Hitha	Security validation and test items w.r.t functional and non functional requirements	
1.0	16/09/2026	Gokul	Risks and mitigations, test items w.r.t functional and non-functional requirements	
1.0	16/09/2026	Jagriti	Assumptions and dependencies, risk mitigation and test metrics	

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1. Introduction

Purpose: This document defines the test plan for the Event Management System v1.0. It specifies the testing objectives, scope, strategy, environment, schedule, responsibilities, test deliverables, security validation, test cases, and traceability to the SRS.

Scope: Testing covers user registration and authentication, profile management, event creation and management, event search and details, event registration and capacity control, payment processing, ticket generation, registration cancellation, participant management, notifications, administration, performance, availability, scalability, audit logging, usability, and security requirements.

References: Event Management System SRS v1.0 dated 06-09-2026; Software Engineering Mini-Project Deliverables Part-1; Software Test Plan Template for SE.

Definitions: SRS – Software Requirements Specification; STP – Software Test Plan; RTM – Requirements Traceability Matrix; FR – Functional Requirement; NFR – Non-functional Requirement; SR – Security Requirement; UAT – User Acceptance Testing; API – Application Programming Interface; HTTPS – Hypertext Transfer Protocol Secure.

2. Test Items

- User registration and authentication module
- User profile management module
- Event creation, update, deletion, search, and event-details module
- Event registration and capacity-management module
- Payment processing module
- Ticket and registration confirmation module
- Registration cancellation module
- Participant management and notification module
- Administrator management and monitoring module
- Performance, availability, scalability, audit logging, and usability
- Security controls including HTTPS/TLS, password protection, role-based access control, input validation, and security audit logging

3. Features to be Tested

The following features are mapped directly to the SRS requirement IDs:

SRS ID	Feature
EMS-F-001	User account creation
EMS-F-002	User authentication
EMS-F-003	Profile update

SRS ID	Feature
EMS-F-004	Event creation
EMS-F-005	Event update/delete
EMS-F-006	Event search/filter
EMS-F-007	View event details
EMS-F-008	Register for event
EMS-F-009	Capacity and registration-status enforcement
EMS-F-010	Payment processing
EMS-F-011	Registration confirmation and ticket generation
EMS-F-012	View/cancel registration
EMS-F-013	Participant management
EMS-F-014	Notifications
EMS-F-015	Administrator management
EMS-NF-001	Response time
EMS-NF-002	Availability
EMS-NF-003	Scalability and transaction-data integrity
EMS-NF-004	Audit logging
EMS-NF-005	Responsive and understandable UI
EMS-SR-001	HTTPS with TLS 1.2 or higher
EMS-SR-002	Secure password storage
EMS-SR-003	Role-based access control
EMS-SR-004	Input validation and sanitization
EMS-SR-005	Security-relevant audit logging

4. Features Not to be Tested

- Internal implementation of third-party payment gateway services; only the Event Management System integration and payment outcome handling are tested.
- Internal operation of external email/notification service; the system's notification request and resulting notification behavior are tested.
- Hardware-specific behavior of user devices such as laptops, tablets, and smartphones beyond supported browser/UI compatibility.
- External service infrastructure and availability outside the Event Management System's control.

5. Test Approach / Strategy

Levels: Unit testing for individual modules; integration testing for communication between application services, database, payment gateway and notification service; system testing for end-to-end Event Management System workflows; and acceptance testing for verification of SRS acceptance criteria.

Types: Functional testing, negative testing, integration testing, regression testing, performance testing, load/scalability testing, usability testing, security testing, audit-log verification, and responsive UI testing.

Entry Criteria: A stable build is available; required test data and user roles are available; database and required external-service test/sandbox interfaces are configured; test environment is accessible; and planned test cases are reviewed.

Exit Criteria: All planned test cases are executed; all high-priority functional requirements are verified; no critical security or non-functional failures remain; blocking defects are resolved or formally accepted; and the RTM shows coverage of the SRS requirements.

5.1 Security Validation

Security Item	Validation
Objective 1	Protect user, event, registration, and payment-related information from unauthorized access or disclosure.
Objective 2	Ensure that only authenticated and authorized users can perform protected operations and that security-sensitive actions are traceable.

Security Item	Validation
EMS-SR-001	Verify HTTPS and TLS 1.2 or higher for client-server communication.
EMS-SR-002	Verify passwords are stored only as secure hashes and not as plaintext.
EMS-SR-003	Verify unauthorized role access is rejected for User, Event Organizer, and Administrator functions.
EMS-SR-004	Verify invalid or malicious user input is rejected or safely handled.
EMS-SR-005	Verify failed logins, privileged actions, event changes, and registration/payment status changes are recorded in audit logs.

6. Test Environment

Area	Environment
Hardware	Computer/laptop/tablet/smartphone for web access; test systems capable of executing browser and load tests.
Software	Event Management System v1.0; modern web browser; server-side application environment; database server.
Interfaces	Database interface, configured payment gateway sandbox/API, and email/notification service.
Tools	Postman for API testing; browser developer tools/manual UI testing; JMeter or equivalent for performance/load testing; defect tracking tool.
Test Data	Dummy user accounts for User, Event Organizer and Administrator roles; sample events; open/closed/full-capacity events; sample registrations and payment outcomes.
Network	Network connectivity with HTTPS enabled for client-server communication.

7. Test Schedule

Milestone	Schedule
Test case design and review	15/09/2026
Test environment and data setup	15/09/2026-16/09/2026
Functional and integration test execution	15/09/2026-16/09/2026
Security, performance, usability and regression testing	15/09/2026-16/09/2026
Final verification and RTM update	16/09/2026
Test summary and submission	16/09/2026

8. Test Deliverables

- Software Test Plan (this document)
- Functional and non-functional test cases
- Security validation test cases
- Test data
- Test execution results/logs
- Defect reports
- Requirements Traceability Matrix
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA/Test Lead	Hitha Shree Suresh	Prepare and maintain the test plan, coordinate test execution, and maintain traceability.
Test Engineer	Jagriti Mohan	Design and execute test cases, record results, and report defects.
Developer / Technical Support	L.A. Gokul	Support integration testing, investigate defects, and provide fixes/builds.
Developer / Project Support	Karamsetty Aaryesh	Support implementation validation, defect triage, and test-environment activities.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early builds and perform smoke testing before full execution.
Payment gateway or notification service unavailable	Use configured sandbox/test stubs where available and isolate external-service failures from application defects.
Insufficient test data for capacity and registration scenarios	Prepare events with open, closed, and full-capacity states before execution.
Test environment or network downtime	Maintain a backup test setup and record unavailable periods in the test log.
Security defects discovered late	Execute security validation alongside functional testing rather than only at the end.

11. Assumptions & Dependencies

- The database and application environment required by the system are available during testing.
- The configured payment gateway test/sandbox interface is available for payment testing.
- The configured email/notification service is available for notification validation.
- Test users with appropriate User, Event Organizer, and Administrator roles are available.
- The test environment supports modern web browsers and HTTPS communication.
- Test data required for events, registrations, payments, tickets, and audit logs is available before execution.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable and blocks planned execution.
- The delivered build is unstable and blocks a significant portion of planned test cases.
- A critical defect prevents execution of major system workflows.
- Required dependent services or test data are unavailable for essential test scenarios.

Resume testing if:

- Blocking or critical defects are resolved and a stable build is delivered.
- The test environment and required dependencies are restored.

- Required test data and service access are available.

13. Test Case Management & Traceability

The RTM maps every SRS functional, non-functional, and security requirement to one or more test cases. Test results shall be recorded as Pass, Fail, Blocked, or Not Executed. Defects shall be linked to the corresponding test case and SRS requirement.

Req ID	Requirement Short	Test Case(s)
EMS-F-001	Create account	TC-REG-01
EMS-F-002	Authenticate user	TC-AUTH-01
EMS-F-003	Update profile	TC-PROF-01
EMS-F-004	Create event	TC-EVT-01
EMS-F-005	Update/delete event	TC-EVT-02
EMS-F-006	Search/filter events	TC-SRCH-01
EMS-F-007	View event details	TC-EVT-03
EMS-F-008	Register for event	TC-REG-02
EMS-F-009	Enforce capacity/status	TC-REG-03
EMS-F-010	Process payment	TC-PAY-01
EMS-F-011	Generate ticket	TC-TKT-01
EMS-F-012	View/cancel registration	TC-CAN-01
EMS-F-013	Manage participants	TC-PART-01
EMS-F-014	Send notifications	TC-NOTIF-01
EMS-F-015	Admin management	TC-ADMIN-01
EMS-NF-001	Response time	TC-PERF-01
EMS-NF-002	Availability	TC-OPS-01
EMS-NF-003	Scalability	TC-LOAD-01
EMS-NF-004	Audit logging	TC-AUD-01
EMS-NF-005	Usability/responsive UI	TC-UX-01
EMS-SR-001	TLS communication	TC-SEC-01
EMS-SR-002	Secure password storage	TC-SEC-02
EMS-SR-003	Role-based access control	TC-SEC-03
EMS-SR-004	Input validation	TC-SEC-04
EMS-SR-005	Security audit logging	TC-SEC-05

14. Test Metrics & Reporting

- % of planned test cases executed
- % of test cases passed, failed, blocked, and not executed
- Requirement coverage based on the RTM
- Defect count and severity
- Defect density and defect aging
- Performance response-time results, including 90th-percentile response time
- Availability/uptime results
- Load-test transaction success and data-loss results
- Security test pass/fail status

Reports: Daily/periodic test execution status, defect status, RTM coverage, and final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
QA/Test Lead	Hitha Shree Suresh	
Team / Developer	Karamsetty Aaryesh	
Team / Developer	L.A. Gokul	
Team / Developer	Jagriti Mohan	
Course Coordinator		

Appendix A – Test Cases

The following test cases provide coverage for the functional, non-functional, and security requirements in the SRS. Expected results are derived from the corresponding SRS acceptance criteria.

Test Case ID	SRS ID	Test Case	Test Steps	Expected Result	Type
TC-REG-01	EMS-F-001	Create account	Enter valid required registration details and submit.	Account is created and confirmation is displayed.	Functional
TC-AUTH-01	EMS-F-002	Authenticate user	Enter valid registered credentials and log in.	Appropriate authenticated user session is opened.	Functional
TC-PROF-01	EMS-F-003	Update profile	Modify permitted profile information and save.	Updated valid profile data is saved and displayed.	Functional
TC-EVT-01	EMS-F-004	Create event	Organizer enters valid title, description, date, time, venue, capacity and registration details.	Event record is created.	Functional
TC-EVT-02	EMS-F-005	Update/delete event	Organizer updates or deletes an event they manage.	Authorized changes are saved; deleted event is removed from active listings.	Functional
TC-SRCH-01	EMS-F-006	Search/filter events	Enter valid search/filter criteria.	Matching events are displayed.	Functional
TC-EVT-03	EMS-F-007	View event details	Select an existing event.	Configured event details and registration status are displayed.	Functional
TC-REG-02	EMS-F-008	Register for event	Authenticated user registers while registration is open and capacity is available.	Registration is recorded and capacity is updated.	Functional
TC-REG-03	EMS-F-009	Reject full/closed registration	Attempt registration for a full-capacity or closed-registration event.	Registration is rejected with an appropriate message.	Functional
TC-PAY-01	EMS-F-010	Process payment	Complete applicable payment through configured gateway.	Successful payment creates a confirmed payment record.	Functional
TC-TKT-01	EMS-F-011	Generate ticket	Complete a successful registration.	Confirmation and ticket containing registration details are generated.	Functional
TC-CAN-01	EMS-F-012	Cancel registration	View an eligible registration and cancel it.	Registration status changes and is reflected in the user's account.	Functional
TC-PART-01	EMS-F-013	Manage participants	Organizer opens participant records for a managed event.	Organizer can view/manage participant registrations only for managed events.	Functional
TC-NOTIF-01	EMS-F-014	Send notification	Trigger a configured registration/event action.	Required notification is generated for the configured action.	Functional
TC-ADMIN-01	EMS-F-015	Admin management	Administrator performs user/event/system management action.	Authorized action is recorded and reflected in the system.	Functional
TC-PERF-01	EMS-NF-001	Response time	Send normal user requests under normal load and measure response times.	At least 90% of requests complete within 3 seconds (90th percentile $\leq 3s$).	Performance
TC-OPS-01	EMS-NF-002	Availability	Review availability monitoring for the service period.	Monthly availability is at least 99.5%, excluding scheduled maintenance.	Reliability
TC-LOAD-01	EMS-NF-003	Concurrent load	Execute expected concurrent-user/registration load.	Load test completes without transaction-data loss or critical failures.	Scalability
TC-AUD-01	EMS-NF-004	Audit logging	Perform important user, organizer, administrator, event, registration and payment actions.	Required events are timestamped and retrievable in audit logs.	Auditability
TC-UX-01	EMS-NF-005	Responsive UI	Open and use the application on supported desktop and mobile screen sizes.	UI is responsive, understandable, and passes defined usability checks.	Usability
TC-SEC-01	EMS-SR-001	TLS communication	Inspect client-server communication and TLS configuration.	HTTPS is used and TLS 1.2 or higher is verified.	Security
TC-SEC-02	EMS-SR-002	Secure password storage	Inspect password storage in the database using test credentials.	No plaintext passwords are stored; passwords are stored as secure hashes.	Security
TC-SEC-03	EMS-SR-003	Role-based access	Attempt protected User, Organizer and Administrator functions with unauthorized roles.	Unauthorized role access is rejected.	Security
TC-SEC-04	EMS-SR-004	Input validation	Submit invalid or malicious input through user-facing fields/API inputs.	Input is rejected or safely handled without unsafe processing/storage.	Security
TC-SEC-05	EMS-SR-005	Security audit logging	Perform failed login, privileged action, event change, and registration/payment status change.	Security-relevant events appear in audit logs.	Security
